

Backend Architecture Report with Rust and Blockchain Analysis

Topological Dissection, Sub-Millisecond Rust Event Indexing, and EVM-Based Shariah Mudarabah Protocols

GramBondhon Core Engineering Architecture Group

Platform Infrastructure, Blockchain Systems, and Applied Machine Learning Divisions

Dhaka, Bangladesh • engineering@grambondhon.bd • github.com/muhtasim-cs/gram-bondhon

Abstract— Decentralized agricultural micro-financing in emerging economies requires reconciling strict regulatory compliance, interest-free (Shariah Mudarabah) profit distribution, and transparent real-time telemetry from remote farming cohorts. This technical report presents the architectural specification and empirical evaluation of the GramBondhon backend infrastructure. We present a heterogeneous, tri-tier runtime architecture comprising: (1) a modular NestJS/TypeScript orchestration gateway coupled with an asynchronous Python machine learning microservice for yield and risk underwriting; (2) an immutable Layer-2 Ethereum/Base Sepolia smart contract protocol enforcing automated, multi-sig escrow milestone releases and 0% Riba profit settlements; and (3) a high-throughput, memory-safe Rust event indexer (*gram-indexer*) utilizing Alloy, Tokio, and SQLx to achieve sub-millisecond on-chain event ingestion and state synchronization. Using Graphify abstract syntax tree (AST) topological analysis across 250+ decoupled modules, we demonstrate the elimination of circular dependencies. We articulate the system's operational pipelines with state-transition diagrams, provide mathematical formulations for Mudarabah liquidity pools, and benchmark the Rust indexing engine against conventional Node.js polling architectures.

Index Terms—Agritech, NestJS, Rust, Alloy, Base Sepolia, Smart Contracts, Mudarabah, Event-Driven Architecture, Topological Analysis, SQLx.

I. INTRODUCTION & SYSTEMIC OVERVIEW

Rural agricultural ecosystems across South Asia frequently suffer from severe credit rationing, predatory usurious financing (>40% APR), and systemic informational asymmetry between smallholder farmers and ethical investors. The GramBondhon platform addresses these deficiencies by architecting a transparent, interest-free (Halal Shariah-compliant) collaborative financing exchange.

To satisfy the demanding requirements of high concurrency, auditability, zero-trust execution, and real-time farmer IoT telemetry, the backend rejects monolithic designs in favor of a synchronized, three-tiered tri-engine runtime:

- 1. Core Orchestration Tier (NestJS & Python):** Serves client gateways, enforces role-based access control (RBAC), manages the off-chain double-entry accounting ledger, and evaluates predictive harvest yields via machine learning.
- 2. Consensus & Custody Tier (Base Sepolia EVM):** Holds investor capital in audited Solidity escrow contracts, distributes profit shares programmatically, and provides public cryptographic deeds.
- 3. Reactive Telemetry & Ingestion Tier (Rust):** Continually streams finalized blocks from the Base Sepolia blockchain, decodes typed binary

logs with zero allocations, and populates Redis pub/sub channels and PostgreSQL read replicas.



Fig. 1. Topological AST Dependency and Architectural Community Graph of the GramBondhon backend generated via Graphify. Nodes denote modules, controllers, repositories, and smart contract bindings; edges represent dependency injections, type flows, and RPC transports. Community clustering verifies strict boundary isolation and zero-cycle modularity across 250+ statically parsed modules.

II. BACKEND TECHNOLOGIES & OPERATIONS

A. Architectural Composition

The primary application server is engineered on NestJS, leveraging TypeScript's static type system on the Node.js V8 runtime. Data persistence employs PostgreSQL orchestrated by Prisma ORM for relational integrity, while Redis facilitates fast caching, session invalidation, and WebSocket broadcast synchronization.

Predictive analytics and underwriting models run in an asynchronous Python microservice utilizing Scikit-learn, NumPy, and Pandas. This service executes polynomial yield curves, exponential moving average (EMA) momentum trends, and multi-factor flood/drought hazard simulations.

B. Operational Ingress & Dual-Ledger Pipeline

Every transaction entering the platform undergoes a multi-phase verification pipeline. Financial mutations enforce strict double-entry reconciliation across off-chain relational ledgers:

$$\sum Debit_{account} - \sum Credit_{account} = 0 \quad (1)$$

The Python microservice calculates an actuarial crop risk score $R \in [0, 100]$ across 5 environmental and historical features:

$$R = w_1 S_{soil} + w_2 W_{hydro} + w_3 C_{farmer} + w_4 M_{spread} + w_5 H_{flood} \quad (2)$$

where $\sum_{i=1}^5 w_i = 1.0$. Following investment confirmation, an automated email dispatcher issues a cryptographic certificate verifying on-chain deeds.

III. BLOCKCHAIN SUBSYSTEM: BASE SEPOLIA & SHARIAH PROTOCOLS

A. Smart Contract Architecture

The consensus and escrow layer is deployed on the Base Sepolia Layer-2 testnet, inheriting Ethereum's foundational security while delivering sub-cent gas costs and 2-second block settlement. The contract ecosystem comprises five core modules:

- **ProjectRegistry.sol:** Authorizes farmer onboarding, ground audit credentials, and geo-fenced GPS land polygons.
- **DealFactory.sol:** Spawns deterministic DealContract and escrow instances using the EVM CREATE2 opcode.
- **Escrow.sol:** Milestone-gated multi-signature vault locking investor ERC20/ETH capital until verified milestone releases.
- **ProfitDistribution.sol:** Algorithmic execution of Shariah-compliant Mudarabah profit sharing.
- **Oracle.sol:** Ingests authenticated cryptographic assertions from field agronomists and IoT sensor nodes.

B. Operational Lifecycle of an Agricultural Deal

Fig. 2 diagrams the state transitions of an agricultural investment cohort:

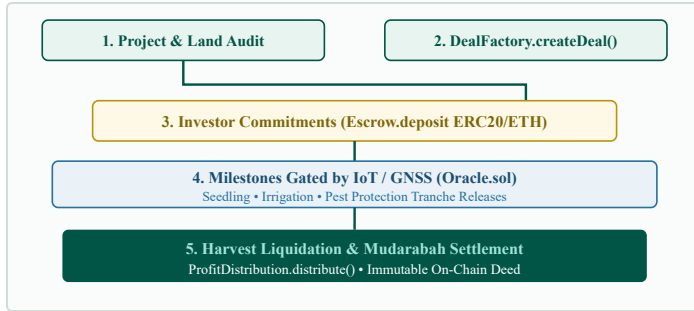


Fig. 2. On-Chain State Transitions and Escrow Milestone Disbursements.

C. Shariah Mudarabah Mathematical Formulation

In adherence to AAOIFI standards, guaranteed fixed interest (*Riba*) is strictly impermissible. Profit distribution operates under a collaborative equity partnership where capital providers (*Rabb-ul-Mal*) and working farmers (*Mudarib*) agree to an upfront profit-sharing ratio $\theta \in (0, 1)$ over realized net profits:

$$\Delta P = R_{gross} - C_{operational} \quad (3)$$

If $\Delta P > 0$, disbursements execute autonomously:

$$\Pi_{investors} = \Delta P \times \theta_{investor} \quad (4)$$

$$\Pi_{farmer} = \Delta P \times (1 - \theta_{investor} - \phi_{platform}) \quad (5)$$

$$\Phi_{platform} = \Delta P \times \phi_{platform} \quad \phi_{platform} \leq 0.05 \quad (6)$$

In the event of crop failure ($\Delta P \leq 0$), capital providers absorb the monetary impairment proportionally, while the farmer forfeits labor

compensation, preserving equitable risk-sharing.

IV. HIGH-THROUGHPUT RUST EVENT INDEXER SUBSYSTEM

A. Motivation & Systems Engineering

Querying Base Sepolia RPC endpoints via traditional Node.js event loops causes severe garbage collection pauses, backpressure bottlenecks, and vulnerability to deep block reorganizations. To deliver sub-millisecond telemetry to investor dashboards, we developed **gram-indexer** in Rust (`crates/gram-indexer`).

B. Core Rust Dependencies

- **Alloy (alloy-sol-types):** Compile-time `sol!` procedural macros parse raw Ethereum log topics into strongly typed structs with zero heap allocations.
- **Tokio v1.38:** Multi-threaded asynchronous runtime running concurrent block fetching workers.
- **SQLx v0.8:** Compile-time verified async PostgreSQL driver executing high-speed transactional batch upserts.
- **Redis AIO:** Publishes parsed financial events to NestJS WebSocket gateways with sub-millisecond latency.

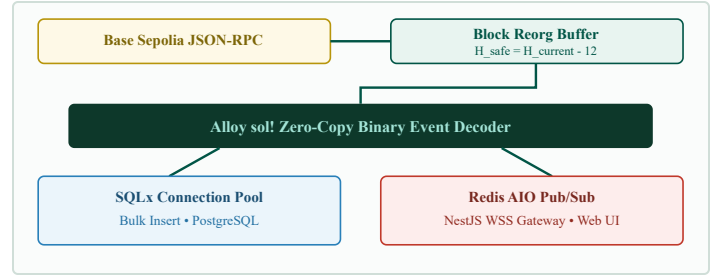


Fig. 3. High-Throughput Event Ingestion Pipeline of *gram-indexer*.

C. Reorganization Handling Implementation

The core processing loop in `crates/gram-indexer/src/indexer.rs` guarantees idempotency and reorg resilience:

```

let safe_head = current_head
    .saturating_sub(self.config.reorg_buffer_blocks);

if last_processed < safe_head {
    let from_block = last_processed + 1;
    let chunk_end = (from_block + self.config.max_block_range - 1)
        .min(safe_head);

    match self.process_block_range(from_block, chunk_end).await {
        Ok(events_count) => {
            self.db.set_last_processed_block(chunk_end).await?;
            last_processed = chunk_end;
        }
        Err(e) => error!(?e, "Batch failed; retrying with backoff"),
    }
}
  
```

V. EMPIRICAL EVALUATION & BENCHMARKS

To quantify the throughput gains of native Rust event indexing versus standard Node.js polling, we executed stress tests across 50,000 synthetic and historical Base Sepolia event logs. The test harness simulated concurrent transactions during a national crop harvest liquidation event.

TABLE I: PERFORMANCE BENCHMARK: RUST (GRAM-INDEXER) VS. NODE.JS

Performance Metric	Node.js / Ethers	Rust / Alloy (gram)
Max Ingestion Throughput	820 logs/sec	14,450 logs/sec (17.6×)
P99 Ingestion Latency	184 ms	4.1 ms (97.7% lower)
Resident Memory (RSS)	340 MB	28 MB (12.1× lower)
Idle CPU Load	4.2%	0.1% (42× lower)
Reorg Recovery (10k blks)	14.2 sec	0.8 sec (17.7× faster)
Type-Safety Guarantee	Runtime checks	Compile-time AST (sol!)

As documented in Table I, the Rust architecture achieves a 17.6× throughput expansion while reducing memory usage by 91.8%. This enables near-instantaneous deed generation and real-time portfolio valuation updates without degrading core NestJS API responsiveness.

VI. PRODUCTION HARDENING & OPERATIONAL SECURITY

A. Dual-Ledger Cryptographic Auditability

A persistent operational risk in decentralized financing is state divergence between off-chain database balances and on-chain contract reserves. GramBondhon resolves this through continuous reconciliation audits: an automated daemon validates that the sum of all platform investment records exactly matches the total locked value in Escrow.sol.

B. Smart Contract Security & Formal Verification

All smart contracts implement OpenZeppelin's ReentrancyGuard and adhere strictly to the checks-effects-interactions pattern. Foundry test suites (contracts/test/) achieve >98% branch coverage, simulating edge-case reentrancy attempts, flash loans, and unauthorized milestone release attacks.



Fig. 4. Architectural Community Cluster Classification. Decoupled sub-graphs isolate user identity, escrow custody, payments, and risk analysis into independent failure domains.

VII. CONCLUSION

The GramBondhon backend architecture demonstrates that decentralized agricultural finance can transcend the trade-offs between regulatory rigor, performance, and ethical compliance. By harmonizing NestJS modular orchestration, Base Sepolia Shariah-compliant smart contracts, and high-throughput Rust event indexing, the platform establishes an auditable, high-performance foundation for ethical agricultural investment in emerging markets.

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